

AMRITSAR, India

WMO: 420710

Lat: 31.710N

Lon: 74.797E

Elev: 230

StdP: 98.59

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.4	3.9	0.8	4.1	7.5	1.9	4.5	8.6	6.1	15.1	5.3	16.0	0.2	270	0.376

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	12.7	42.2	23.0	40.9	23.0	39.2	22.9	30.0	32.2	29.2	32.5	28.8	32.4	3.4	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.8	27.7	30.8	28.9	26.1	30.5	28.0	24.8	30.7	102.9	30.8	98.5	32.4	95.9	33.3	34.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.8	5.5	4.7	0.4	45.1	1.1	1.1	-0.4	45.9	-1.0	46.5	-1.6	47.2	-2.4	48.0		
			0.2	30.7	1.2	1.1	-0.6	31.4	-1.3	32.0	-2.0	32.6	-2.9	33.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.3	11.1	14.6	19.6	26.0	31.1	32.0	30.3	29.7	28.2	24.7	18.5	12.8
	DBStd	7.66	2.27	2.48	3.02	3.27	2.60	2.83	2.02	1.71	1.65	2.39	2.24	2.35
	HDD10.0	21	15	1	0	0	0	0	0	0	0	0	0	4
	HDD18.3	549	225	106	21	0	0	0	0	0	0	0	25	172
	CDD10.0	4859	48	130	298	479	653	661	630	612	547	454	254	92
	CDD18.3	2346	0	2	61	229	395	411	372	354	297	196	29	0
	CDH23.3	31754	1	37	711	2967	5991	6344	5172	4662	3488	1979	391	11
CDH26.7	17739	0	1	232	1718	3930	4138	2816	2341	1636	864	63	0	
Wind	WSAvg	1.4	1.0	1.4	1.6	2.0	2.2	2.2	1.5	1.1	1.0	0.8	0.7	0.7
Precipitation	PrecAvg	673	26	37	30	16	19	70	193	180	89	9	6	9
	PrecMax	1023	79	119	115	59	64	232	458	458	417	62	48	45
	PrecMin	410	0	0	0	0	0	0	58	11	3	0	0	0
	PrecStd	182	22	37	31	17	17	52	99	114	85	17	10	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.9	26.1	33.2	41.5	44.2	44.2	39.9	36.5	36.0	34.2	30.0	24.9
		MCWB	14.1	17.7	20.9	20.6	22.0	23.5	26.6	28.3	26.2	24.2	20.1	15.2
	2%	DB	20.1	24.2	31.1	39.2	42.2	42.9	37.8	35.4	35.0	33.2	28.4	23.0
		MCWB	13.7	16.7	20.6	20.9	22.5	23.5	27.3	28.1	26.4	23.9	19.1	15.1
	5%	DB	19.0	22.9	29.1	37.2	40.9	41.1	36.1	34.8	34.0	32.3	27.2	21.4
		MCWB	13.3	16.1	19.7	20.4	22.3	24.1	27.8	28.0	26.6	23.3	18.5	14.5
	10%	DB	17.5	21.1	27.2	35.0	39.2	39.4	34.9	33.9	33.1	31.2	26.0	20.0
		MCWB	12.6	15.0	18.9	20.1	22.0	24.0	27.7	27.9	26.3	22.8	17.6	13.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.2	19.5	23.7	24.2	27.3	29.6	31.2	31.1	29.5	28.2	22.1	17.7
		MCDB	19.1	23.1	28.6	34.7	38.3	33.4	32.5	31.9	31.6	30.9	27.4	21.1
	2%	WB	15.2	18.2	21.9	22.7	25.8	28.7	30.1	30.0	28.6	26.3	20.8	16.3
		MCDB	17.8	22.3	28.5	33.3	36.7	34.3	32.2	32.1	31.7	30.2	26.9	20.9
	5%	WB	14.1	17.1	20.8	21.9	24.7	27.8	29.4	29.2	28.0	25.2	19.6	15.4
		MCDB	17.2	21.1	27.4	32.2	35.8	34.1	32.5	32.2	31.5	30.0	25.5	19.8
	10%	WB	13.1	16.0	19.7	21.1	23.8	27.1	28.8	28.7	27.3	24.1	18.5	14.3
		MCDB	16.4	20.0	26.0	31.5	34.9	33.9	32.6	32.0	31.1	29.2	24.2	18.7
Mean Daily Temperature Range	5% DB	MDBR	12.4	13.0	14.1	16.2	15.6	12.7	7.8	7.4	9.5	13.2	15.3	14.0
		MCDBR	15.0	15.0	16.2	19.0	17.5	15.9	9.8	8.6	10.6	13.4	16.3	16.4
		MCWBR	9.2	8.2	7.3	5.5	5.3	4.6	4.0	3.7	4.7	5.6	8.1	9.4
	5% WB	MCDBR	12.9	12.8	14.7	16.7	15.8	11.2	7.8	7.5	8.8	11.2	15.0	14.7
		MCWBR	9.0	8.1	7.4	6.1	6.6	5.4	4.2	4.1	4.7	5.6	8.6	9.7
Clear-Sky Solar Irradiance	taub	0.483	0.494	0.510	0.566	0.646	0.737	0.776	0.686	0.596	0.639	0.597	0.527	
	taud	1.863	1.888	1.883	1.801	1.665	1.543	1.518	1.677	1.811	1.629	1.657	1.776	
	Ebn at Noon	699	747	773	748	693	629	603	652	692	608	583	624	
	Edh at Noon	166	178	192	216	250	281	288	242	204	228	204	174	
All-Sky Solar Radiation	RadAvg	2.87	3.90	5.24	6.28	6.97	6.51	5.42	5.20	5.20	4.67	3.60	2.97	
	RadStd	0.32	0.43	0.35	0.29	0.25	0.40	0.40	0.41	0.28	0.25	0.19	0.19	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page